

Attention's Substates: Unlocking Adaptive VR through EEG Insights

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ABSTRACT

To enable adaptive virtual reality (VR) based systems, this study explored to differentiate attention's 3 substates – i.e., orienting (OR), alerting (AL), and cognitive maintenance (CM). Human participants performed a task within a VR-based cockpit to evoke these substates, while their brain activity was recorded as electroencephalographic data to extract features. Statistical analyses revealed significant differences of these features among the substates. Compared 5 machine-learning models, naïve bayes and K-nearest neighbors achieved similarly the best accuracy (77% ~ 95%) to classify the features. These findings implied a potential of the features for informing design of adaptive VR-based systems.

Keywords: VR, attention substates, EEG, machine learning

Index Terms: Human-centered computing ~ Human computer interaction (HCI) ~ Interaction paradigms ~ Virtual reality • Human-centered computing ~ Human computer interaction (HCI) ~ HCI design and evaluation methods ~ User studies

1 INTRODUCTION

To enhance human performance, attention is paramount to inform adaptive design of a training system based on virtual reality (VR) technologies. Attention is a cognitive state that allows a human user to dynamically respond to information. While being considered as a binary value (i.e., yes or no) traditionally, attention can be divided into 3 interconnected substates: orienting (OR), alerting (AL), and cognitive maintenance (CM) [1]. OR directs the user to a specific stimulus; AL prepares the user's sensitivity to the stimulus; and CM sustains the user's focus over time. The operation of these substates could like a dynamic flow, adapting to real-time tasks. For example, a pilot must maintain focus on controlling an aircraft (CM), while redirecting the cognition to external stimuli (OR) and staying prepared for unexpected events (AL). However, the dynamic flow among the substates remains underexplored when the user executes such a task in VR-based systems.

To understand the relationship between attention and its substates, a feasible approach is to acquire and analyse brain activity during a VR-based task. Electroencephalography (EEG) would be a suitable tool for this acquisition. With its high temporal resolution, EEG data can contain rapid transitions between substates of a cognitive state (e.g., attention) [2]. It is known that certain loci of the cerebral cortex (the brain) are associated with attention [3]. Thus, it is possible to place EEG electrodes on the scalp overlapping the loci to acquire brain activity associate with attention and its substates [1 - 3]. OR is linked to the loci of Fp1, Fp2, Fz, T3, T4, T5, T6, Pz, P3, and P4 – according to the 10-20 electrode-placement standard [2] – of the dorsal attention network. AL is related to the ventral attention network, overlapping the loci

of Fp1, Fp2, Fz, F3, F4, T3, T4, T5, T6, and Pz. CM involves sustained activation of the loci including Fp1, Fp2, Fz, F3, F4, T5, T6, Pz, P3, P4, C3 and C4. All these loci enable to map attention onto its substates during the execution of the task.

To analyse EEG data of the brain activity, machine-learning (ML) models are a conventional instrument for classifying the binary value of attention [4]. Examples of the models are naïve bayes, random forest (RF), K-nearest neighbours (KNN), support vector machines (SVM), and gradient boosting (GB). On one hand, tuning hyperparameters of these models is a difficult endeavour. On another hand, the models have not been used to classify the substates of attention from the EEG data, leaving a gap to hinder adaptive design of VR-based training systems.

To address this gap, we conducted a feasibility study on differentiating the substates of attention from EEG data. Within a VR-based cockpit, the quasi-practical task of a human user (i.e., acting as a pilot) was to land an aircraft. While landing, the user needed to count the number of red lights on a matrix mimicking a precision approach path indicator (PAPI) – a standard part of the landing procedure. We acquired EEG data of the user while executing the task and compared the 5 ML models – as aforementioned – for classifying the 3 substates of attention. We found that it is feasible to differentiate the substates from the EEG data. Statistical analyses of extracted features from the EEG data confirmed significant differences among the substates. Moreover, Naïve bayes achieved the best accuracy of the classification. These findings highlighted a potential of applying the substates to adaptive design of VR-based systems.

2 METHODS

Study Setup: As illustrated in Fig. 1, the visual scene of an aircraft's cockpit was created by using Unity (2022.3.9f). Sitting in the cockpit, each participant (a pseudo pilot) could see a forest and a lake, along with a PAPI-style light matrix overlaying on the upper-right corner of the cockpit. The scene was displayed on a 3D autostereoscopic monitor (Acer Spatial Labs). Employing the right hand, the participant controlled the aircraft by manipulating the stylus of a TOUCH haptic device (3D Systems). However, no force feedback was provided to the hand, alike an actual cockpit. The participant wore a 20-electrode Enobias EEG cap (Neuroelectrics). EEG data were recorded at a sampling rate of 500 Hz.

The setup above was running on a graphic computer (Dell Precision 5820), integrating multithreaded processes for visual



Figure 1: Study setup - the visual scene of a VR-based cockpit (left) and the layout of the setup with a human participant facing an autostereoscopic monitor, wearing an EEG cap, and articulating the stylus of a haptic device (right).

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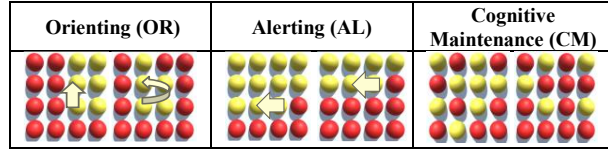


Figure 2: Red lights on a PAPI-style matrix for attention.

rendering, haptic interfacing, EEG recording, signal processing, and power spectral density (PSD) computation.

Participants: Unaware of the study's purpose, 16 participants (8 males and 8 females, 28.7 ± 2.4 years old) took part in the study. A pre-screening confirmed them with right-handedness, normal colour vision, stereo acuity of at least $40''$ of arc, and no physical impairment for right-hand movements. None of them had a history of attention deficiency. The study had an ethics clearance from our institute. All participants gave informed consents.

Procedure: The task of each participant had 3 types, and each type was designated to evoke a specific substate of attention. As shown in Fig. 2, the lights on the PAPI-style matrix illuminated in vertical (or counter-clock) directions for the OR type, requiring the participant to shift his/her focus dynamically along the directions. The same lights activated in right-to-left sequences for the AL type, prompting him/her to anticipate the sequences. The lights turned on and off at random locations for the CM type, challenging him/her to sustain the focus over time. Each trial was one of these 3 types. At the end of the trial, the participant reported the counted number of the red lights.

Sitting comfortably, each participant held the stylus while looking at the autostereoscopic monitor. While staring at a cross on the monitor, the participant underwent the protocol of eye-closed-eye-opening for recording a 3.0-minute baseline of his/her EEG data. Then, a learning session of 10 trials ensured the participant to familiarize with the task. Afterwards, a testing session of 30 trials (10 trials per type) was carried out. The order of the 30 trials was randomized. The recording of EEG data continued during both the learning and testing sessions. The participant spent about 45.0 minutes in the study, including breaks.

Data Analyses: A pipeline for analysing EEG data was implemented to classify the 3 substates. Preprocessing of the data used a bandpass filter (1.0 – 59.0 Hz) to exclude powerline effect and an independent component analysis to remove common artifacts like eye movements. Then, the data was segmented into trial-specific epochs. The PSD of each epoch was computed using the Welch method with a 2.0-second window and 50% overlap. Thus, EEG features extracted from the data were the PSD values of each frequency band (i.e., the δ , θ , α , β , and γ bands [3]) and of each electrode (i.e., 20). Afterwards, these features were used in the following two steps of analyses.

The first step was ANOVA to check whether the 3 substates of attention were statistically differentiable. For each participant, the feature average of a frequency band was calculated among all electrodes and epochs associated with a substate. ANOVA was applied to the averages of each band among all participants, with the Kolmogorov-Smirnov (KS) test for normality.

The second step involved comparing the five ML models for classifying the substates. The EEG features of each participant were organized into a 30×100 matrix, with a row representing a trial and a column corresponding to one of 100 features (= 5 bands $\times$ 20 electrodes = one set). Each set was scaled using the z-score normalization. The matrices were split into a training (80%) and a testing (20%) subgroup with gender-balanced stratification. The ML models were trained and tested on the matrices. The

Table 1: Results of the KS tests for all frequency bands

Bands	δ	θ	α	β	γ
KS	0.49; $p = 0.027$	0.31; $p = 0.034$	0.64; $p = 0.052$	0.25; $p = 0.005$	0.68; $p = 0.056$

Table 2: Classification accuracy of the 5 ML models

	Naïve Bayes	RF	KNN	SVM	GB
OR	91%	89%	78%	91%	68%
AL	85%	86%	95%	72%	85%
CM	77%	74%	89%	57%	94%

classification performance of the models was evaluated based on accuracy, referring to the designated task types for all epochs.

3 RESULTS AND DISCUSSION

The EEG data of all participants was used to extract PSD features for the statistical analyses and the comparison of the ML models. As shown in Table 1, the KS tests confirmed the normality of the feature averages of each frequency band. An ANOVA indicated a significant difference of the feature averages among all frequency bands [$F(4, 495) = 173.61$; $p < 0.01$]. Another ANOVA also revealed a significant difference among the 3 substates [$F(2, 87) = 3.10$; $p < 0.05$], confirming their distinct patterns of brain activity. This revelation underlined the ML classification of the substates.

As indicated in Table 2, naïve bayes and KNN performed the best among all 5 ML models. This performance might correspond to their approaches, which could capture distributional differences in the EEG features. However, these 2 models exhibited varying preferences for each substate. OR exhibited a better accuracy of 91.0% for naïve bayes than of 78% for KNN. This might be a consequence of rapid shifts to stimuli, creating highly distinct probabilistic patterns to favor naïve bayes. AL had an accuracy of 85% for naïve bayes and of 95% for KNN, resulting possibly from pattern variability of tracking stimuli's sequences to appease KNN. With its relatively undominant patterns, CM had an accuracy of 77.0% for naïve bayes and of 89.0% for KNN to prefer KNN, reflecting sustained focus to cause undominant patterns. Indeed, future work needs to refine EEG features for better differentiation among the substates of attention, including more participants for increasing observation robustness and applying Bonferroni or Holm-Sidak corrections for enhancing statistical rigor.

4 CONCLUSION

This study confirmed the feasibility of using the conventional ML models, especially naïve bayes and KNN, for differentiating the 3 substates of attention. These findings could pave a way of informing design of adaptive VR-based systems.

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